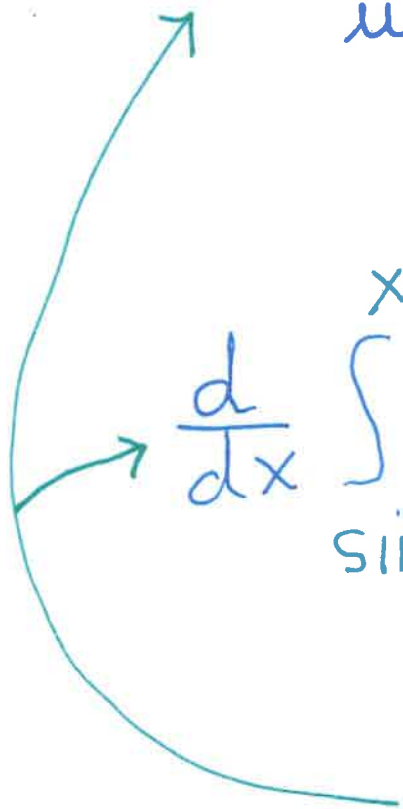


①

$$\frac{d}{dx} \int_{u(x)}^{v(x)} f(t) dt = f(v(x))v'(x) - f(u(x))u'(x)$$


$$\begin{aligned} \frac{d}{dx} \int_{\sin x}^{x^3} e^{t^2} dt &= e^{(x^3)^2} \cdot 3x^2 - e^{(\sin x)^2} \cdot \cos x \\ &= e^{x^6} \cdot 3x^2 - e^{\sin^2 x} \cdot \cos x \end{aligned}$$

From Last Time

②

Integration by Substitution

Chain Rule $\frac{d}{dx} F(g(x)) = F'(g(x))g'(x)$

$$\int F'(g(x))g'(x) dx = F(g(x)) + C$$

$$\text{Put } f(x) = F'(x)$$

$$\int f(g(x))g'(x) dx = F(g(x)) + C$$

$$\text{Put } u = g(x), \frac{du}{dx} = g'(x), du = g'(x)dx$$

$$\int f(u) du = F(u) + C$$

$$\text{Ex: } \int x^2(x^3+9)^5 dx$$

$$= \frac{1}{3} \int u^5 du$$

$$= \frac{1}{3} \left(\frac{1}{6} u^6 \right) + C$$

$$= \frac{1}{18} (x^3+9)^6 + C$$

③

$$u = x^3 + 9$$

$$du = 3x^2 dx$$

$$x^2 dx = \frac{1}{3} du$$

$$\text{Ex: } \int x \sqrt{x-1} \, dx$$

(4)

$$= \int (u+1) u^{1/2} \, du$$

$$= \int u^{3/2} + u^{1/2} \, du$$

$$= \frac{2}{5} u^{5/2} + \frac{2}{3} u^{3/2} + C$$

$$= \frac{2}{5} (x-1)^{5/2} + \frac{2}{3} (x-1)^{3/2} + C$$

$$\begin{aligned} u &= \sqrt{x-1} \\ du &= \frac{dx}{2\sqrt{x-1}} \\ ?? \end{aligned}$$

$$\begin{aligned} u &= x-1 \\ du &= dx \\ x &= u+1 \end{aligned}$$

Integration by Parts

⑤

$$\frac{d}{dx} (f(x)g(x)) = f(x)g'(x) + f'(x)g(x)$$

$$f(x)g'(x) = \frac{d}{dx} (f(x)g(x)) - f'(x)g(x)$$

$$\int f(x)g'(x)dx = f(x)g(x) - \int g(x)f'(x)dx$$

$$u = f(x)$$

$$du = f'(x)dx$$

$$dv = g'(x)dx$$

$$v = g(x)$$

$$\int u dv = uv - \int v du$$

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Ex: $\int x e^x dx$

$$= x e^x - \int e^x dx$$

$$= x e^x - e^x + C$$

$$\begin{aligned} u &= x \\ du &= dx \\ dv &= e^x dx \\ v &= e^x \end{aligned}$$

ILATE $\leftarrow$ Exponential

Log $\nearrow$ $\nwarrow$ Trig

$\nwarrow$ Inverse $\swarrow$ Algebraic

Farther to left, better choice for u

⑦

Ex: $\int \ln x \, dx$

$$= x \ln x - \int dx$$
$$= x \ln x - x + C$$

$$u = \ln x$$

$$du = \frac{dx}{x}$$

$$dv = dx$$

$$v = x$$

⑧

Ex: f is diff, $f'(x) > 0$ all $x \in [0, 1]$

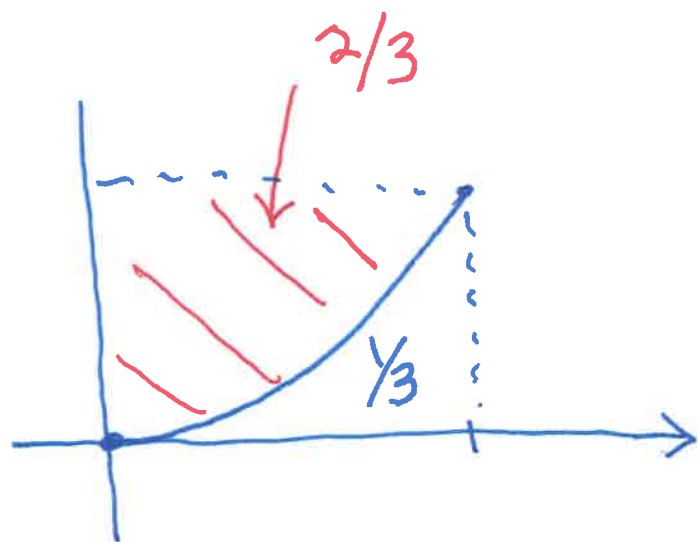
$$f(0) = 0, \quad f(1) = 1, \quad \int_0^1 f(x) dx = \frac{1}{3}$$

Find $\int_0^1 f^{-1}(y) dy$

$$f^{-1}(f(x)) = x, \quad x \in [0, 1]$$

$$\text{Claim: } \int_0^1 f^{-1}(y) dy = \frac{2}{3}$$

9



$$\int_0^1 f^{-1}(y) dy = \frac{2}{3}$$

$$\begin{aligned} \int_0^1 f^{-1}(y) dy & \quad \swarrow \text{parts} \\ &= \int_0^1 x f'(x) dx \\ &= x f(x) \Big|_0^1 - \int_0^1 f(x) dx \\ &= 1 - 0 - \frac{1}{3} = \frac{2}{3} \end{aligned}$$

Analytical Soln

Subs. $y = f(x)$

$$x = f^{-1}(y)$$

$$dy = f'(x) dx$$

$$x=0 \leftrightarrow y=0$$

$$x=1 \leftrightarrow y=1$$

(10)

Improper Integrals

$$\int_I f(x) dx$$

interval $\nearrow$

Improper if either

(i) I is unbounded or

(ii) f blows up on I

(or both)

②

Focus on Case where

- (a) I is unbounded
- (b) f is bounded on every bounded subinterval of I
- (c) f has at most a finite # of disc. on each bounded subinterval of I .

(12)

Ex: $\int_1^{\infty} \frac{dx}{x^3}$ Fix $t > 1$

Look at $\int_1^t \frac{dx}{x^3}$, then let $t \rightarrow \infty$

$$\begin{aligned} \int_1^t \frac{dx}{x^3} &= \int_1^t x^{-3} dx = -\frac{1}{2} x^{-2} \Big|_1^t \\ &= -\frac{1}{2t^2} + \frac{1}{2} \end{aligned}$$

$$\lim_{t \rightarrow \infty} \int_1^t \frac{dx}{x^3} = \frac{1}{2}$$

(13)

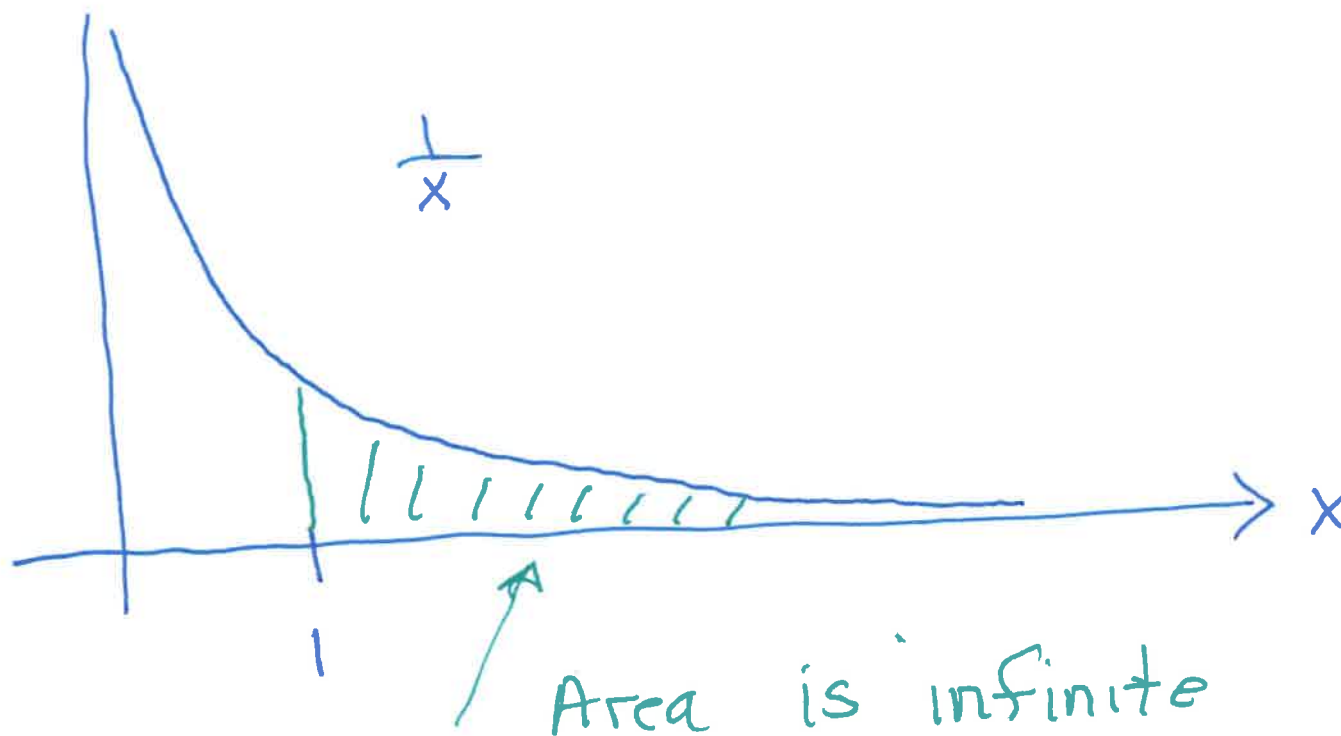
Integral Converges

$$\int_1^{\infty} \frac{dx}{x^3} = \frac{1}{2}$$

Ex: $\int_1^t \frac{dx}{x} = \ln x \Big|_1^t = \ln t$

$$\lim_{t \rightarrow \infty} \int_1^t \frac{dx}{x} = \infty \quad \int_1^{\infty} \frac{dx}{x} \text{ Diverges}$$

(14)



(15)

Def: If $\lim_{t \rightarrow \infty} \int_a^t f(x) dx$ exists and

is finite we say that

$\int_a^{\infty} f(x) dx$ converges and we

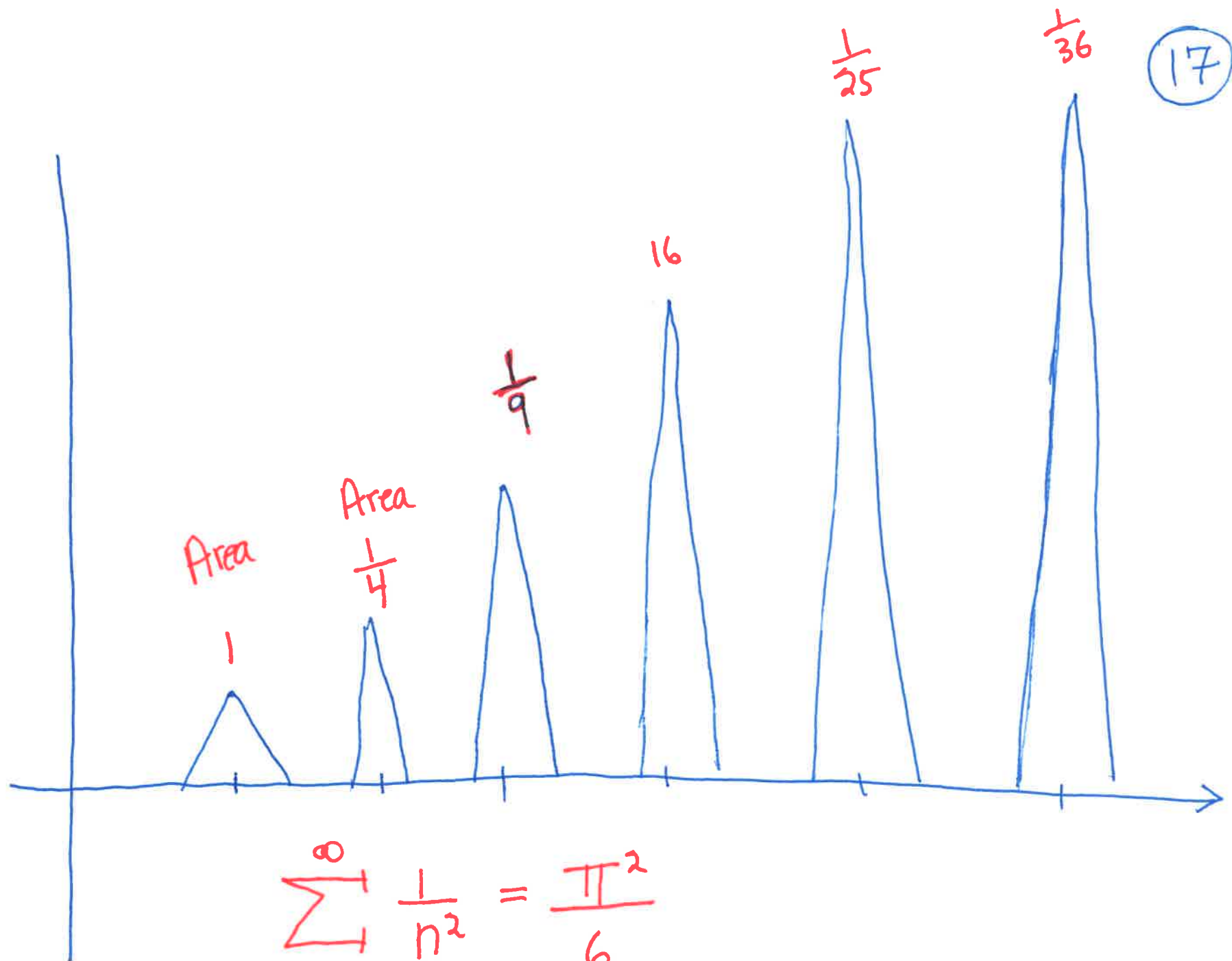
write $\int_a^{\infty} f(x) dx = \lim_{t \rightarrow \infty} \int_a^t f(x) dx$

(16)

Question: Assume f is cont. on $[0, \infty)$,
that $f(x) \geq 0$ for all $x \geq 0$, and
that $\int_0^{\infty} f(x) dx$ converges.

Can we conclude that
 $f(x) \rightarrow 0$ as $x \rightarrow \infty$

Answer: NO



(18)

Similarly for $\int_{-\infty}^b f(x) dx$

Look at $\lim_{t \rightarrow \infty} \int_{-t}^b f(x) dx$

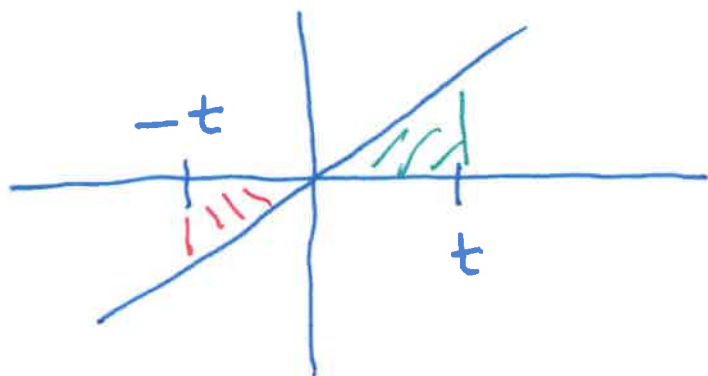
For $\int_{-\infty}^{\infty} f(x) dx$, we can't just

look at $\lim_{t \rightarrow \infty} \int_{-t}^t f(x) dx$

(19)

$$\int_{-t}^t x dx = 0$$

$\int_{-\infty}^{\infty} x dx$ does not
Converge



$\int_{-\infty}^{\infty} f(x) dx$ Converges means that

$\int_{-\infty}^0 f(x) dx$ Conv and $\int_0^{\infty} f(x) dx$ Conv.

$$\int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^0 f(x) dx + \int_0^{\infty} f(x) dx$$

(20)

Comparison Test: Assume that $f: [a, \infty) \rightarrow \mathbb{R}$ and $g: [c, \infty) \rightarrow \mathbb{R}$ are bounded and that $d \geq \max\{a, c\}$. Assume also that

$$0 \leq g(x) \leq f(x) \text{ for all } x \geq d.$$

(a) If $\int_a^{\infty} f(x) dx$ converges then so does $\int_c^{\infty} g(x) dx$.

(b) If $\int_c^{\infty} g(x) dx$ diverges then so does $\int_a^{\infty} f(x) dx$.

(21)

Using fact that f is bdd on $[a, d]$ and g is bounded on $[c, d]$

$$\int_a^{\infty} f(x) dx \text{ Conv} \iff \int_d^{\infty} f(x) dx \text{ Conv.}$$

$$\int_c^{\infty} g(x) dx \text{ Conv} \iff \int_d^{\infty} g(x) dx \text{ Conv.}$$

(22)

Important Comparison Fns

I. Claim $\int_1^{\infty} \frac{dx}{x^p}$ converges if
and only if $p > 1$

(Exercise)

II. Claim $\int_a^{\infty} e^{-\alpha x} dx, \alpha > 0$
always converges

(23)

$$\int_a^t e^{-\alpha x} dx = -\frac{1}{\alpha} e^{-\alpha x} \Big|_a^t$$

$$= -\frac{1}{\alpha} (e^{-\alpha t} - e^{-\alpha a})$$

$$\rightarrow \frac{e^{-\alpha a}}{\alpha} \quad \text{as } t \rightarrow \infty$$

$$\int_a^{\infty} e^{-\alpha x} dx = \frac{1}{\alpha} e^{-\alpha a}$$

(24)

Ex: $\int_0^{\infty} e^{-x^2} dx$ Converges

$$e^{-x^2} \leq e^{-x}, \quad \text{for } x \geq 1$$

$$\int_0^{\infty} e^{-x^2} dx = \int_0^1 e^{-x^2} dx + \int_1^{\infty} e^{-x^2} dx$$

$\int_1^{\infty} e^{-x^2} dx$ Conv. by Comp Test

(25)

Ex: $\int_0^{\infty} \frac{dx}{1+x^4}$

Converges

$$\frac{1}{1+x^4} \leq \frac{1}{x^4}$$

$$\int_1^{\infty} \frac{dx}{x^4}$$

Conv.

$$\frac{1}{1+x^4} \text{ bdd on } [0,1]$$

$$\int_0^{\infty} \frac{dx}{1+x^4}$$

Conv.

(26)

Ex: $\int_0^{\infty} \frac{dx}{\sqrt{1+x^2}}$

Diverges

$$\frac{1}{\sqrt{1+x^2}} \sim \frac{1}{x}$$

for x large

$$\frac{1}{\sqrt{1+x^2}} \leq \frac{1}{x}$$

$$x \geq 1$$

$$1 \leq x^2 \text{ for } x \geq 1$$

$$\sqrt{1+x^2} \leq \sqrt{x^2+x^2}, \quad x \geq 1$$

$$\sqrt{1+x^2} \leq \sqrt{2} x, \quad x \geq 1$$

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$$\frac{1}{\sqrt{1+x^2}} \geq \frac{1}{\sqrt{2}} \left(\frac{1}{x} \right) \quad x \geq 1$$

$$\int_1^{\infty} \frac{dx}{x} \quad \text{Diverges}$$

$$\int_0^{\infty} \frac{dx}{\sqrt{1+x^2}} \quad \text{Diverges}$$

What Would you do with

$$\int_0^{\infty} x^{100} e^{-x^{1/3}} dx ?$$

$$= \int_0^{\infty} y^{300} \cdot 3y^2 e^{-y} dy$$

$$= 3 \int_0^{\infty} y^{302} e^{-y} dy$$

$$y = x^{1/3}$$

$$y = 0 \text{ when } x = 0$$

$$y \rightarrow \infty \text{ when } x \rightarrow \infty$$

$$x = y^3, \quad dx = 3y^2 dy$$

Exponentials

CRUSH

Polynomials

Fact: Given any $p, q > 0$ there exist constants $C, a > 0$ s.t.

$$e^{-(x^p)} \leq \frac{C}{x^q} \quad \text{all } x \geq a.$$

(30)

Ex: Choose $c, a > 0$ s.t.

$$e^{-x^{1/3}} \leq \frac{c}{x^{102}}, \quad x \geq a$$

(We are using $p = \frac{1}{3}$ and $q = 102$)

$$x^{100} e^{-x^{1/3}} \leq x^{100} \left(\frac{c}{x^{102}} \right) = \frac{c}{x^2}, \quad x \geq a$$

$\int_a^{\infty} \frac{dx}{x^2}$ Converges, So

$\int_0^{\infty} x^{100} e^{-x^{1/3}} dx$ Converges.